

Worksheet 4 *All about linear models*Prepares you for lab manual pages **D1, D2, D3, RT1, RT2** (section L2 S2)

Name:

Name of TA:

1. Suppose that F is a quantity that depends on time t . Assume that $\Delta F / \Delta t = 2$ (constant!) and $F(0) = 3$.(a) Draw a time series for the quantity F .(b) What is $F(2)$?(c) What is $F(4)$?(d) What is ΔF from $t = 2$ to $t = 4$?(e) What is ΔF on the time interval from $t = 2$ to $t = 2.01$?(f) What is the general (predictive) formula for the function $F(t)$? $F(t) =$ **2.** Write down five different functions $F(t)$ that satisfy the change equation $\Delta F / \Delta t = -3$. (For example, $F(t) = t^2$ is a function, and $F(t) = 2 + 7t$ is a function.)**3.** In each of the following, give the change equation $\Delta F / \Delta t = \underline{\hspace{2cm}}$ satisfied by the function.

(a) $F(t) = 3t + 7$

$\Delta F / \Delta t =$

(b) $F(t) = 9t - 2$

$\Delta F / \Delta t =$

(c) $F(t) = 1 - 3t$

$\Delta F / \Delta t =$

(d) $F(t) = 5t - 2t$

$\Delta F / \Delta t =$

4. If K is the number of kittens, t is measured in days, and $K = 3t + 8$, what is $\Delta K / \Delta t$? Include the appropriate units.